

Graphical models for infinite measures

with applications to Extremes and Lévy processes

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Bochum - 14 August 2024

Joint work with Sebastian Engelke and Jevgenijs Ivanovs



1. Motivation

Multivariate extremes & Graphical modelling

2. Bridging/extending existing frameworks?

New notion of conditional independence
for (not only) exponent measures

3. Results for undirected and directed graphs

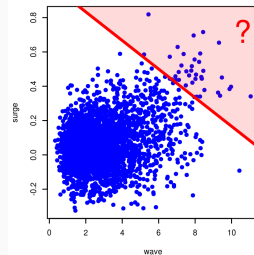
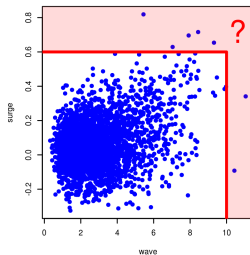
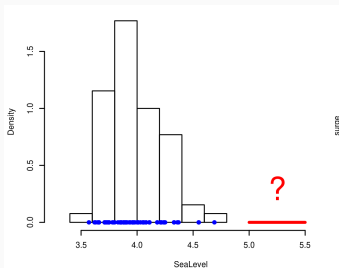
4. Summary

<https://doi.org/10.48550/arXiv.2211.15769>

Motivation

Extreme value theory

= theoretical basis for **extrapolation into the regime of rare events**



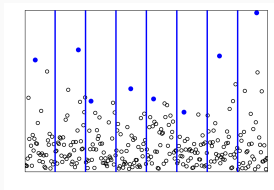
data portpirie and wavesurge
from R package ismev

Typical applications

- hydrology (e.g. dikes), meteorology (wind gusts, heavy rainfall)
- finance/insurance (reserve funds to cover big losses)

Univariate "vanilla" approaches

Consider repeated measurements of 1 variable (e.g. temperature at 1 location)

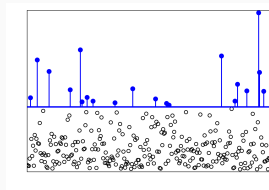


Block maxima $(M_j)_{j=1,2,\dots,r}$
 $\approx$ Sample from **max-stable distribution** (GEV)

$$G_{\gamma}\left(\frac{x - \mu}{\sigma}\right)$$

with $G_{\gamma}(x) = \exp\{-(1 + \gamma x)_+^{-1/\gamma}\}$

(Fisher-Tippet-Gnedenko)



Threshold-Exceedances $X_j | X_j > t$
 $\approx$ Sample from **threshold-stable distribution** (GPD)

$$H_{\gamma}\left(\frac{x}{\sigma}\right)$$

with $H_{\gamma}(x) = \{1 - (1 + \gamma x)_+^{-1/\gamma}\}_+$

(Pickands-Balkema-de Haan)

A point processes view

With $(X_k) \stackrel{\text{i.i.d.}}{\sim} X \sim F$, $\bar{\Lambda} = -\log G$, $H = (1 + \log G)_+$, $a_n = \sigma(b_n)$, $b_n = F^{\leftarrow}(1 - 1/n)$, the following are **equivalent**:

Poisson Point Process limit

$$\sum_{k=1}^n \delta_{k/n, (X_k - b_n)/a_n} \xrightarrow{\mathcal{D}} \text{PPP}(\text{Leb} \times \Lambda)$$

Max-stable limit (GEV)

$$\frac{\max_{k=1}^n X_k - b_n}{a_n} \xrightarrow{\mathcal{D}} G$$

Threshold-stable limit (GPD)

$$\frac{X - u}{\sigma(u)} \Big|_{X > u} \xrightarrow{\mathcal{D}} H$$



A. Bücher and C. Zhou.

A horse racing between the block maxima method and the peak-over-threshold approach.

Statistical Science 36(3):360-378, 2021.

From univariate to **multivariate** extremes

With $(X_k) \stackrel{\text{i.i.d.}}{\sim} X \geq 0 \in \mathbb{R}^d$ and $a_n = F_{\|X\|_\infty}^{\leftarrow}(1 - 1/n)$, the following are equivalent¹:

Poisson Point Process limit

$$\sum_{k=1}^n \delta_{(k/n, X_k/a_n)} \xrightarrow{\mathcal{D}} \text{PPP}(\text{Leb} \times \Lambda)$$

Max-stable limit

$$\frac{\max_{k=1}^n X_k}{a_n} \xrightarrow{\mathcal{D}} G$$

$$G(x) = \exp \left[-\Lambda([0, \infty)^d \setminus [0, x]) \right]$$

Threshold-stable limit

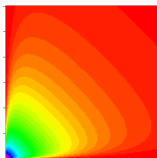
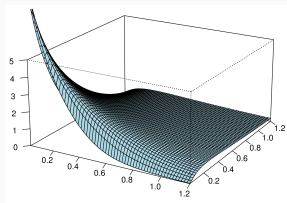
$$\frac{X}{u} \mid \|X\|_\infty > u \xrightarrow{\mathcal{D}} H$$

$$H(dy) = \frac{\Lambda(dy)}{\Lambda(L)}, \quad L = [0, \infty)^d \setminus [0, 1]^d$$

¹e.g. nicely summarised in [Dombry/Ribatet Stat Interface (2015): Functional regular variations, Pareto processes and peaks over threshold]

Properties of the exponent measure Λ

- **finite**
away from the origin $0 \in \mathbb{R}^d$
- **homogeneous**: $\Lambda(tA) = t^{-\alpha}\Lambda(A)$
- **explodes** at 0, but $\Lambda(\{0\}) = 0$
- can be seen as **Lévy-measure** wrt **max**:
 $X \sim G_\Lambda$ is **max-infinite divisible**,
 $X \sim \max_{i=1}^\infty \eta_i$, where $\{\eta_i\} \sim \text{PPP}(\Lambda)$.



- Generalizations:**
- Lévy-measures wrt other (semi-)group operations
[Davydov et al. EJP (2008): Strictly stable distr. on convex cones]
 - limiting measures for regularly varying $X \in \mathbb{R}^d$

Graphical models for extremes?

Why? – sparsity, factorizations, efficiency gains, interpretation, ...

First failure: impossible/unnatural? GPSD 2016 (in Bochum!):

Kirstin Strokorb¹, Ioannis Papastathopoulos²

¹University of Mannheim, Mannheim

²University of Edinburgh, Edinburgh

Conditional independence among max-stable laws

02.03.2016

S3 | 14:00

HZO70

Independence and conditional independence are key concepts in the theory of probability and statistical inference. In general, neither does independence imply conditional independence nor does conditional independence imply independence of the subvectors of a random vector. However, we show that, if the distribution of a *max-stable* random vector has a *positive continuous density*, then conditional independence of any two subvectors conditioned on the remaining components implies already their independence. We conclude that a broad class of commonly used tractable max-stable models cannot exhibit an interesting Markov structure. This result also complements the recent developments in Dombry and Eyi-Minko (2014), where it is shown that, up to time reversal, only max-autoregressive processes of order one can appear as discrete time stationary max-stable Markov chains.

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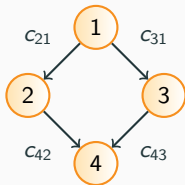
Better ideas:

- "top down" approaches: next slide
- "bottom up" approaches: J. Segers + co-authors

Two recent approaches to graphical modelling for Extremes

Max-linear approach

[Klüppelberg/Gissibl/Lauritzen/Tran/Améndola/...]



- considers **directed acyclic graphs**

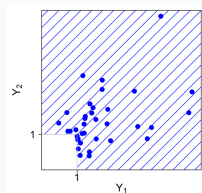
$$X_i = \bigvee_{j \in \text{pa}(i)} c_{ij} X_j \vee Z_i, \quad i \in V$$

with with indep. $Z_i > 0$, $c_{ij} > 0$.

- links to tropical geometry
- for max-stable distr.: **restricted to spectrally discrete models**

Threshold approach

[Engelke/Hitz JRSSB Discussion 2020]



- CI notion related to **density factorization** on $[0, \infty)^d \setminus [0, 1]^d$

$$h(y) = \frac{h_{A \cup C}(y_{A \cup C}) h_{B \cup C}(y_{B \cup C})}{h_C(y_C)}$$

- Graphs **necessarily connected**
- forbids** (asymptotic) **independent models**

Bridging theory
via **exponent measures?**

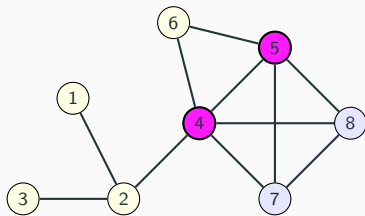
Goal:
Graphical models for $\wedge$

Graphical models **are based on** conditional independence

- Write $X_A = (X_\alpha)_{\alpha \in A}$ and abbreviate $X_A \perp\!\!\!\perp X_B \mid X_C$ by $A \perp\!\!\!\perp B \mid C$.
- Graph $\mathcal{G} = (V, E)$ (with direction) gives meaning to (non-)adjacency, separation, (non-)descendence, neighbours, parents etc

Markov properties	Undirected	Directed
Pairwise	for non-adj. (α, β) : $\alpha \perp\!\!\!\perp \beta \mid V \setminus \{\alpha, \beta\}$	for non-adj. (α, β) with $\beta \in \text{nd}(\alpha)$: $\alpha \perp\!\!\!\perp \beta \mid \text{nd}(\alpha) \setminus \{\beta\}$
Local	for any α : $\alpha \perp\!\!\!\perp V \setminus (\{\alpha\} \cup \text{ne}(\alpha)) \mid \text{ne}(\alpha)$	for any α : $\alpha \perp\!\!\!\perp (\text{nd}(\alpha) \setminus \text{p}(\alpha)) \mid \text{p}(\alpha)$
Global	if S sep. A from B : $A \perp\!\!\!\perp B \mid S$	if S dir.-sep. A from B : $A \perp\!\!\!\perp B \mid S$

Example



If $(X_i)_{i=1,\dots,8}$ is
(undirected) globally Markov
wrt this graph:

$$\{1, 2, 3, 6\} \perp\!\!\!\perp \{7, 8\} \mid \{4, 5\}$$

$$\{6\} \perp\!\!\!\perp \{8\} \mid \{4, 5\}$$

$$\{1\} \perp\!\!\!\perp \{7\} \mid \{4\}$$

...

General approach:
Conditional independence wrt Λ

A natural definition of conditional independence

- Setting:
- Partition (A, B, C) of the index set $V = \{1, \dots, d\}$
 - Exponent (or other) measure Λ on $\mathbb{R}^d$

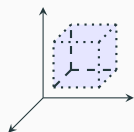
Definition (Conditional independence wrt Λ)

$$A \perp B \mid C \quad [\Lambda]$$

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Test sets:



- charged product from sets R bd. away from the origin
- Probability law $\mathcal{L}_R$

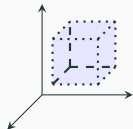
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$$A \perp B \mid C \quad [\Lambda] \quad \text{if} \quad Y_A \perp\!\!\!\perp Y_B \mid Y_C \quad [Y \sim \mathcal{L}_R] \quad \text{for any } R \in \mathcal{R}.$$

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“Every feasible product form restriction of Λ results in (classical) conditional independence.”

Notation & Explosiveness assumptions

Let $\emptyset \neq D \subset V = \{1, 2, \dots, d\}$.

- **Marginal measure:** $\Lambda_D(\cdot) = \Lambda(\{y : y_D \in \cdot\})$
- **Restricted measure:** $\Lambda_D^0(\cdot) = \Lambda(\{y : y_D \in \cdot, y_{V \setminus D} = 0_{V \setminus D}\})$

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-

Explosiveness assumptions:

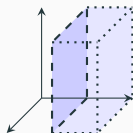
- (E0) **Explosion:** $\Lambda(y_v \neq 0) \in \{0, \infty\}$ for all $v \in V$
- (E1) **Explosion in all sub-faces:** (E0) for Λ_D^0 for all $d \in D \subset V$

(✓satisfied for homogeneous measures)

Alternative characterizations of $A \perp B \mid C \ [\Lambda]$

- **Reduced test classes:** Sufficient to check defn only for **test sets**

$$R_{v,\varepsilon} = \{y \in \mathbb{R}^d : |y_v| > \varepsilon\}$$



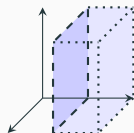
1. for all $v \in V$, $\varepsilon > 0$
2. for all $v \in C$, $\varepsilon > 0$ **and** $A \perp B \ [\Lambda_{A \cup B}^0]$

→ retrieve [Engelke/Hitz 2020 JRSSB] approach
(including probabilistic interpretation among multivariate Pareto models)

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→ retrieve [Engelke/Hitz 2020 JRSSB] approach
(including probabilistic interpretation among multivariate Pareto models)

- **Density factorization** wrt product measure μ (if density exists):

$$\lambda(y)\lambda_C(y_C) = \lambda_{A \cup B}(y_{A \cup B})\lambda_{B \cup C}(y_{B \cup C})$$

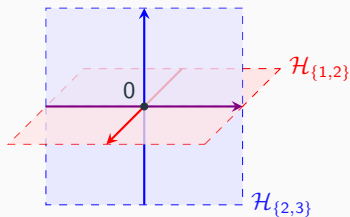
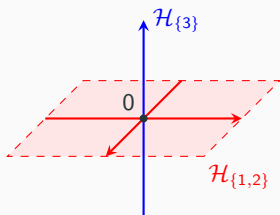
for μ -almost all y with $y_C \neq 0_C$ **and** $A \perp B \ [\Lambda_{A \cup B}^0]$.

Independence

Proposition (Independence)

Assume (E0). Then

$$A \perp B \quad [\wedge] \quad \text{iff} \quad \Lambda(\{y_A \neq 0_A \text{ and } y_B \neq 0_B\}) = 0.$$



Mass concentrating on $\mathcal{H}$ -spaces:

Left: $\{1, 2\} \perp \{3\} \quad [\wedge]$; **Right:** $\{1\} \perp \{3\} \quad [\wedge]$.

Modified densities lead to factorization on restricted spaces

Assume (E1) and Λ with density wrt

$$\mu = \bigotimes_{v \in V} \mu_v \quad \text{such that} \quad \mu_v(\{0_v\}) > 0 \text{ for all } v \in V.$$

Definition: **Modified density** for $\emptyset \neq D \subset V$

$$\bar{\lambda}_D(y_D) = \begin{cases} \lambda_D(y_D) & \text{for } y_D \neq 0_D, \\ 1/\mu_D(\{0_D\}) & \text{for } y_D = 0_D. \end{cases}$$

More natural factorization:

$A \perp B \mid C [\Lambda]$ is equivalent to

$$\bar{\lambda}(y) \bar{\lambda}_C(y_C) = \bar{\lambda}_{A \cup C}(y_{A \cup C}) \bar{\lambda}_{B \cup C}(y_{B \cup C})$$

for μ -almost all $y \in \mathbb{R}^d \setminus \{y : y_A \neq 0_A, y_B \neq 0_B, y_C = 0_C\}$.

Semi-graphoid properties [Pearl '88]

Theorem (Semi-graphoid)

Assuming (E1), we have for disjoint subsets A, B, C, D of V

- (L1) (Symmetry) If $A \perp B \mid C [\Lambda]$, then $B \perp A \mid C [\Lambda]$.
- (L2) (Decomposition) If $A \perp (B \cup D) \mid C [\Lambda]$, then $A \perp B \mid C [\Lambda]$.
- (L3) (Weak union) If $A \perp (B \cup D) \mid C [\Lambda]$, then $A \perp B \mid C \cup D [\Lambda]$.
- (L4) (Contraction) If $A \perp B \mid C [\Lambda]$ and $A \perp D \mid B \cup C [\Lambda]$,
then $A \perp (B \cup D) \mid C [\Lambda]$.

Only (L4) requires Assumption (E1).

When do we have a **graphoid**?

When does (L1)-(L4) and

(L5) (*Intersection*) If $A \perp B \mid C \cup D [\Lambda]$ and $A \perp C \mid B \cup D [\Lambda]$,
then $A \perp B \cup C \mid D [\Lambda]$.

hold?

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hold?

Quite subtle!

- In general not!
Easy counterexample: put mass on a ray through 0.
More subtle counterexamples exist.
- The [Engelke/Hitz JRSSB 20] models (and independent concatenations thereof) will do.

So far: Conditional independence wrt Λ

Now: Graphical models wrt Λ ?

Memo: Markov properties wrt graph $\mathcal{G} = (V, E)$

	Undirected	Directed
Pairwise	for non-adj. (α, β) : $\alpha \perp\!\!\!\perp \beta \mid V \setminus \{\alpha, \beta\}$	for non-adj. (α, β) with $\beta \in \text{nd}(\alpha)$: $\alpha \perp\!\!\!\perp \beta \mid \text{nd}(\alpha) \setminus \{\beta\}$
Local	for any α : $\alpha \perp\!\!\!\perp V \setminus (\{\alpha\} \cup \text{ne}(\alpha)) \mid \text{ne}(\alpha)$	for any α : $\alpha \perp\!\!\!\perp (\text{nd}(\alpha) \setminus \text{p}(\alpha)) \mid \text{p}(\alpha)$
Global	if S sep. A from B : $A \perp\!\!\!\perp B \mid S$	if S dir.-sep. A from B : $A \perp\!\!\!\perp B \mid S$

Undirected graphical models wrt Λ

Main result: **Hammersley-Clifford type theorem**

Consider Λ on a **decomposable graph** $\mathcal{G}$ with **modified density** $\bar{\lambda}$ wrt μ .

Theorem

Assume (E1). Then Λ is *globally Markov* if and only if

$$\bar{\lambda}(y) \prod_{S \in \mathcal{S}} \bar{\lambda}_S(y_S) = \prod_{C \in \mathcal{C}} \bar{\lambda}_C(y_C)$$

for μ -almost all $y \notin \mathcal{Z}(\mathcal{G})$.

Here

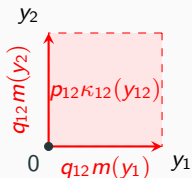
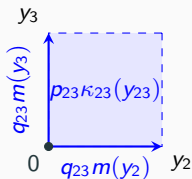
$$\mathcal{Z}(\mathcal{G}) = \bigcup_{\substack{a, b \in V, S \in \mathcal{S}: \\ S \text{ separates } a \text{ from } b}} \{y \in \mathbb{R}^V : y_a \neq 0, y_b \neq 0, y_S = 0_S\}.$$

→ **modified density** is the "natural" object for factorization properties

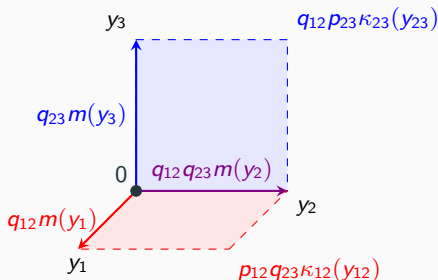
A generic construction

Example for a measure Λ with identical marginals satisfying $\{1\} \perp \{3\} \mid \{2\}$ (globally Markov wrt chain graph $1 - 2 - 3$)

Building blocks



Resulting Λ (densities on faces)



on inner 3-dim'l face:

$$p_{12}p_{23}\kappa_{12}(y_{12})\kappa_{23}(y_{23})/m(y_2)$$

→ beyond [Engelke/Hitz 2020 JRSSB] approach: mass on faces allowed + no homogeneity needed

Directed graphical models wrt Λ

Fundamental observation: **DG = DL**

DG = Directed global Markov property

$A \perp B \mid S \ [\wedge]$ if A and B are separated by S in the moral graph of the smallest ancestral set containing $A \cup B \cup S$.

DL = Directed local Markov property

$\alpha \perp (\text{nd}(\alpha) \setminus \text{p}(\alpha)) \mid \text{p}(\alpha) \ [\wedge]$ for any node α .

Proposition

Assume (E1).

(DG) holds for $\wedge$ iff **(DL)** holds for $\wedge$

Proof. Only needs semi-graphoid properties (L1)-(L4) [Lauritzen et al. 1990].

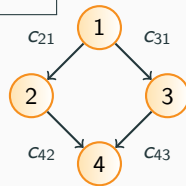
Structural max-linear models are Bayesian networks wrt $\perp_\Lambda$



C. Améndola, C. Klüppelberg, S. Lauritzen, N. Tran
Conditional independence in max-linear Bayesian networks.
Ann. Appl. Probab. 32 (1), 1-45 (2022).

consider **directed acyclic graphs** $G = (V, E)$ with

$$X_i = \bigvee_{j \in \text{pa}(i)} c_{ij} X_j \vee \varepsilon_i, \quad i \in V$$



- **By defn:** X_V satisfies **classical (DL)** property.
- X_V is a max-linear model with **exponent measure**

$$\Lambda(E) = \sum_{\substack{j=1, \dots, q \\ c_{ij}^* \neq 0}} \mu_j(\{t > 0 : tc_{ij}^* \in E\}).$$

Proposition (MLBN are BN for $\perp_\Lambda$)

Λ satisfies **(DL)**. It also satisfies **(DG)** if $\varepsilon_i > 0$ for all i .

Conclusions

- new notion of **conditional independence** $\perp_{\Lambda}$
for measures Λ that can explode at the origin

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for measures on decomposable graphs

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Thank you!

Further novelties (not this talk):

- unified framework for max-id and id vectors; extends also to Lévy processes and beyond
- graphical models exhibiting asymptotic independence

<https://doi.org/10.48550/arXiv.2211.15769>

Key references

- **Threshold approach to CI in Extremes**

S. Engelke and A. Hitz. Graphical models for extremes (with discussion).
J. R. Stat. Soc. Ser. B Stat. Methodol., 82:871932, 2020.

- **Max-linear approach to CI in Extremes**

C. Améndola, C. Klüppelberg, S. Lauritzen, N. M. Tran. Conditional independence in max-linear Bayesian networks. Ann. Appl. Probab. 32(1), 1-45, 2022.

- **Fundamentals of Graphical models**

S. L. Lauritzen. Graphical Models. Oxford University Press, 1996.

Appendix

Independence: $X \perp\!\!\!\perp Y$

$$\begin{aligned} & \mathbf{P}\{X \in A, Y \in B\} \\ &= \mathbf{P}\{X \in A\} \mathbf{P}\{Y \in B\} \end{aligned}$$

$$\mathbf{P}\{X \in A \mid Y\} \stackrel{\text{a.s.}}{=} \mathbf{P}\{X \in A\}$$

Y has no further info on X

density formulation

$$\pi_{XY}(x, y) = \pi_X(x) \pi_Y(y)$$

Conditional independence: $X \perp\!\!\!\perp Y \mid Z$

$$\begin{aligned} & \mathbf{P}\{X \in A, Y \in B \mid Z\} \\ & \stackrel{\text{a.s.}}{=} \mathbf{P}\{X \in A \mid Z\} \mathbf{P}\{Y \in B \mid Z\} \end{aligned}$$

$$\mathbf{P}\{X \in A \mid Y, Z\} \stackrel{\text{a.s.}}{=} \mathbf{P}\{X \in A \mid Z\}$$

*Y has no further info on X
if we know already Z*

density formulation

$$\begin{aligned} & \pi_{XYZ}(x, y, z) \\ &= \pi_{X|Z}(x|z) \pi_{Y|Z}(y|z) \pi_Z(z) \end{aligned}$$

Example: Gaussian Markov random fields (GMRF)

$\mathbf{X} = (X_i)_{i=1,\dots,d}$ Gaussian random vector

Well-known:

Independence

(Σ covariance matrix)

$$X_i \perp\!\!\!\perp X_j \quad \Leftrightarrow \quad \Sigma_{ij} = 0$$

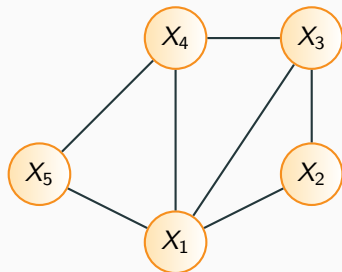
Conditional independence

($Q = \Sigma^{-1}$ precision matrix)

$$X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{-ij} \quad \Leftrightarrow \quad Q_{ij} = 0$$

Graph formulation (GMRF):

No edge if $Q_{ij} = 0$.



Statistical inference:
based on MCMC and
sparse matrix techniques
(very efficient)

cf. [Rue/Held (2005): GMRFs]

Applications - Gallery

Danube - Flow Speed

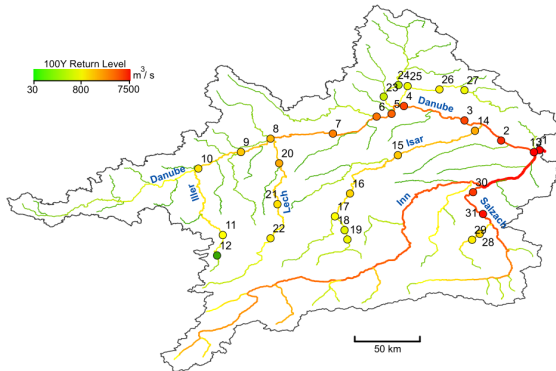
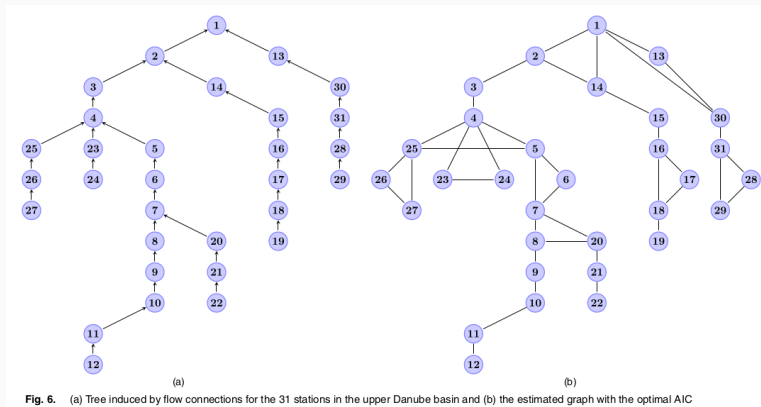


FIG. 6. 100-year return levels for river flow (m^3/s), extrapolated to the entire network T ; the colors of the points indicate the return levels at the 31 numbered gauging stations.

Danube - Flow Speed



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Danube - Flow Speed

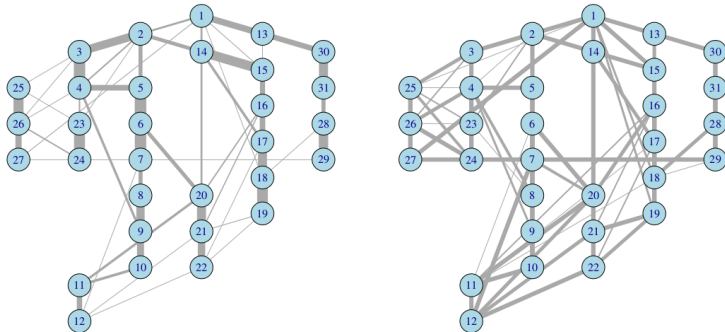


FIGURE 3. Left: Extremal graphical structure of the fitted EMTP_2 Hüsler-Reiss distribution; edge thickness is proportional to $\log(1 - \hat{\Theta}_{ij})$. Right: Summary of estimated graphical structures of the fitted EMTP_2 Hüsler-Reiss distributions for different thresholds $p \in \{0.7, 0.75, 0.8, 0.85, 0.9, 0.95\}$; edge thickness represents the proportion of occurrences of the edge among the estimated graphs.

Danube - Flow Speed

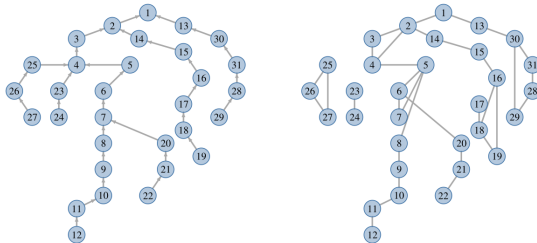


FIG 8. *The physical flow connection tree (left) and the selected graph by MBIC estimated by EGlearn (right) for the Danube data set.*

Engelke/Lalancette/Volgushev - Learning extremal graphical structures in high dimensions

Airport Delays

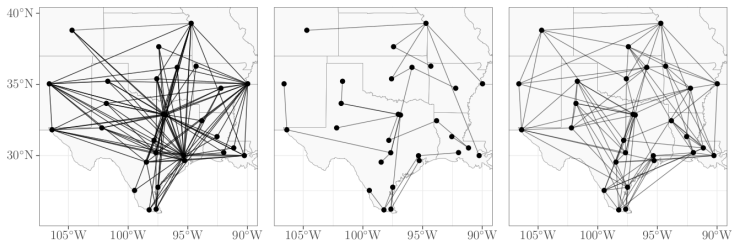


Figure 5: The flight graph G_{flight} (left), the estimated tree graph $\hat{T}$ (center), and the graph estimated using **EGlearn**, $\hat{G}_{\rho^*}$, for $\rho^* = 0.15$ (right).

Airport Delays

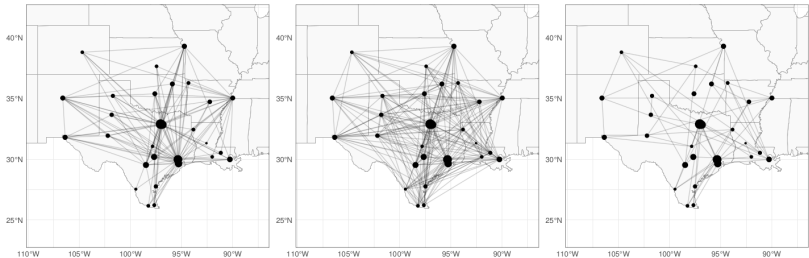


FIGURE 5. Airports in the Southern U.S. (dots) and flight connections, where the thickness of the nodes indicates the average number of daily flights at the airports. Left: flight connection graph with an edge between any pair of airports with daily flights. Center: estimated graph of optimal **eglearn** model. Right: estimated sub-graph corresponding to observed variables of optimal **eglatent** model.

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Currency Exchange Rates

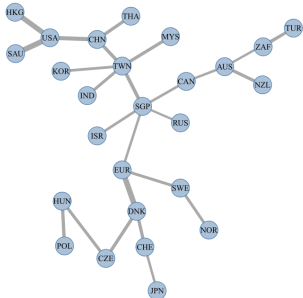


FIGURE 9 Minimum spanning tree T_T of extremal dependence for the spot foreign exchange rate data based on the combined extremal variogram. The width of each edge $(i, j) \in E_T$ is proportional to the extremal correlation $2 - 2\Phi(\sqrt{F_{ij}}/2)$, and therefore wider edges indicate stronger extremal dependence. [Colour figure can be viewed at wileyonlinelibrary.com]

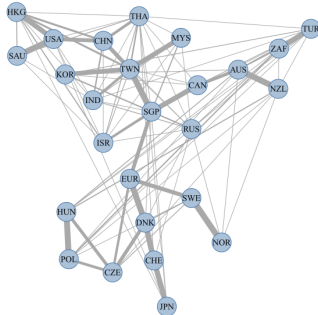


FIGURE 10 Graph where the width of each edge is proportional to the number of times it has been selected in an extremal tree in the bootstrap procedure. [Colour figure can be viewed at wileyonlinelibrary.com]

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- JRSSB 2022

Currency Exchange Rates

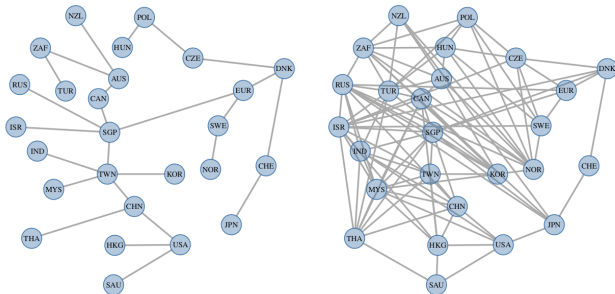


FIG 9. The minimum spanning tree (left) and the selected graph by MBIC estimated by *EGlearn* (right) for the exchange rates data set; the list of currencies with their respective abbreviations are shown in Table S2 in Supplementary Material S1.4.

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Graphical lasso

